SE375 SYSTEM PROGRAMMING SPRING 2025-2026
Laboratory Assignment

March 27, 2026

1. Introduction

Assume that each district (Konak, Gaziemir etc) of İzmir reports their data on public transport usage and the overall data set is collected for analysis. The data set from each district contains the daily number of bus passengers and number of bus trips. An example view of a data set from one district (say Konak) is as following;

Date	Day of the Week	Number of Trips	Number of Passengers
01.02.2023	Wednesday	34	1242
02.02.2023	Thursday	51	4589
03.02.2023	Friday	30	2568
04.02.2023	Saturday	48	3874
05.02.2023	Sunday	36	2514
06.02.2023	Monday	44	6257
07.02.2023	Tuesday	27	2674

2. Task

In this lab assignment, you are going to develop a Java Application that will calculate statistics on given data. Your application will read each data file, compute and store the values and report to the user.

a. Specifications

- Each data file contains the daily number of bus trips and number of bus passengers in a CSV (comma-separated values) format for each district. The first line of the file has the headers in the order of:

  Date, *Day of the Week*, *Number of Trips* and *Number of Passengers*

  and the remaining lines include data for each date of the month.
- `DD.MM.YYYY` is the date format.
- Day of the week is one of the below;

Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday

- The data type of *Number of Trips* and *Number of Passengers* is integer. It is guaranteed that the data files do not have missing values.
- Your application should use two `HashMap` structures:
    - One for the task given in Lab Assignment #2,
    - One for the task given in Lab Assignment #3.
        - If you haven't finished these tasks already, you can access previous lab assignment instructions through Blackboard.
- Your application should **create** a thread pool with size 10.
- Your application should **assign** a thread for each data file from the thread pool and give both the file name and the `HashMap` as an argument. This structure should be shared between all threads.
- Each thread should **read** the given data file.
- Each thread should **update** both `HashMap` structures correctly while processing the given data file.
- Once threads are finished, your application should **display** the data in both `HashMap` structures. The main application should not update any `HashMap` at any point.
- Each thread should be **synchronized** by using `Locks`, instead of using the `synchronized` keyword.
- Implement two versions of this task:
    - One that locks both `HashMaps` at the same time
    - One that locks one `HashMap`, processes it, unlocks it and locks the next one.
    - For both versions, write a piece of code that reports the running time. You can use the code fragment below for this purpose:

```
long startTime = System.nanoTime();
// The operation
long endTime = System.nanoTime();
long elapsedTime = endTime - startTime;
```

    - Run both versions 50 times, find the average running time for each and report them in the comments of your code.

You can check previous assignments' instruction documents in order to check whether your results are correct or not.

Regulations

- You should use Java.
- Late submission is **not** allowed.
- This is an **individual** assignment. Plagiarism, copying, cheating, outsourcing the exam to another person or organization for with or without pay are considered as actions of academic dishonesty. Failure to maintain academic honesty may result in disciplinary action according to the Izmir University of Economics' disciplinary bylaw

for students of institutions of higher education
(https://www.ieu.edu.tr/en/bylaws/type/read/id/13).

- Submissions should be done to Blackboard. Submissions via e-mail will not be accepted.
- You should make sure to obey the specifications.

Submission

- You should submit a zip file to Blackboard containing your Java files.
- By submitting you pledge the following; *"On my honor, as an Izmir University of Economics student, I affirm that I have not given or received any unauthorized help on this exam, and that this work is my own."*